

SHAHWAIZ MEMON

MEng Electrical & Computer Engineering

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PROFESSIONAL SUMMARY

I am a published machine learning researcher and MEng student who enjoys turning open-ended questions into measurable experiments and practical systems. My work is driven by curiosity about how intelligent systems learn, where they fail, and how they can become more efficient and reliable. I am most energized by research that combines scientific exploration with meaningful real-world impact.

EDUCATION

M.Eng. in Electrical and Computer Engineering

Jan 2026 – Present

University of Victoria - Victoria, Canada

- CGPA: 8.0/9.0
- MEng Research: Investigating lightweight ML models for battery state-of-health and remaining-useful-life prediction under computational and hardware constraints.
- Directed Studies: Developing AI-based detection and early prediction of sensor-spoofing attacks on a Raspberry Pi smart thermostat using TON_IoT sensor data.
- Relevant coursework: Optimization for Machine Learning, Data Analysis and Pattern Recognition, Applied Data Analysis, Security, Privacy and Data Analytics, and Design and Analysis of Computer Communication Networks.

B.Sc. in Computer Science

NUCES FAST - Islamabad, Pakistan

Aug 2020 – May 2024

- Dean's List of Honour.
- Relevant coursework: Artificial Intelligence, Generative AI, Data Mining, Statistical Modeling, Data Structures and Algorithms, Software Engineering, Operating Systems, and Computer Architecture.

RESEARCH & PUBLICATIONS

Research Assistant Intern

Data-Insight Lab - Islamabad, Pakistan

Aug 2023 – May 2024

- Developed EcoLLM-Mini within the CaptionCraft project through supervised fine-tuning, knowledge distillation, profiling, and hyperparameter optimization, reducing computational requirements by 88% and accelerating inference by 92%.
- Designed controlled experiments and comparative evaluations across vision-language models, analyzing output quality, inference latency, computational efficiency, and deployment trade-offs.
- Publication: Shahwaiz Memon, et al. "A Light-Weight Image to Description Model Using Knowledge Distillation." IEEE AIBThings 2024.

Research & ML PROJECTS

AI-Based Intrusion Detection for IoT Systems

Directed Studies Project, UVic

Aug 2026 - Present

- Developing a Raspberry Pi smart-thermostat simulation with live sensor-spoofing attacks and training ML classifiers on TON_IoT sensor data for attack detection and early prediction.
- Designing controlled experiments around accuracy, false-positive rate, response time, robustness, and resource-constrained deployment.
- Tools: Python, scikit-learn, Pandas, Raspberry Pi, TON_IoT, Linux

Battery Systems Optimization

MEng Project, UVic

Jan 2026 - Present

- Investigating lightweight ML approaches for battery state-of-health and remaining-useful-life prediction under computational and hardware constraints.
- Benchmarking model accuracy, computational cost, inference latency, and deployment feasibility through systematic experimentation and hyperparameter optimization.
- Tools: Python, MATLAB, PyTorch, Pandas, NumPy, scikit-learn

Automated Product Information Management

Professional Project, Scala Teams

Jan 2025 - Apr 2026

- Developed an end-to-end RAG and multi-agent system enabling natural-language query, creation, and update of structured product data through reasoning, tool use, APIs, and external services.
- Designed prompts, routing logic, guardrails, automated evaluation, tracing, monitoring, and error recovery to analyze agent reliability and failure modes.
- Tools: Python, LangChain, LangGraph, Vertex AI, Google ADK, MCP, FastAPI, MLflow, Grafana

CaptionCraft

Undergraduate Capstone, NUCES FAST

Aug 2023 - May 2024

- Integrated BLIP, BLIP-2, CLIP, and Llama 2 into a PyTorch multimodal image-to-description pipeline and deployed the optimized model in a mobile prototype.
- Applied fine-tuning, knowledge distillation, profiling, validation, and comparative benchmarking to improve model efficiency and evaluate generalization and deployment trade-offs.
- Tools: Python, PyTorch, Hugging Face Transformers, FastAPI, Flutter, Linux, Git, GCP

ACADEMIC EXPERIENCE

Teaching Assistant (Marker & Lab)

University of Victoria

May 2026 – Present

- Supported courses in Data Analysis and Pattern Recognition, Blockchain Technologies, and Software Engineering Methods.
- Facilitated labs, communicated technical concepts, assessed assignments, and provided structured feedback to students with varied experience levels.

RELEVANT PROFESSIONAL EXPERIENCE

Junior Machine Learning Engineer

Scala Teams LLC - Islamabad, Pakistan

Jun 2024 – Apr 2026

- Developed production LLM and multi-agent systems integrating RAG, APIs, external tools, structured data, and workflow state.
- Designed prompts, agent instructions, routing logic, guardrails, and evaluation workflows to analyze agent behaviour, task performance, failure modes, and output quality.
- Built parallelized Python and SQL benchmark infrastructure, reducing model-evaluation time by 53%, and collaborated across teams to debug and productionize experimental AI systems.

Machine Learning Consultant

Upstart Commerce - Islamabad, Pakistan

Aug 2023 – May 2024

- Collaborated with faculty and industry stakeholders to define research questions, datasets, evaluation criteria, and deployment constraints for a generative-AI application.
- Developed and evaluated an end-to-end text-generation pipeline, comparing output quality, inference speed, computational requirements, and deployment efficiency.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, SQL, MATLAB, Bash

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, Deep Learning, Computer Vision, Multimodal Learning, LLMs, Fine-Tuning, Knowledge Distillation

Research Methods: Experiment Design, Training, Validation, Benchmarking, Profiling, Hyperparameter Optimization, Statistical Analysis, Model Evaluation

Agentic AI: RAG, LangChain, LangGraph, Google ADK, Vertex AI, MCP, Tool Calling, Guardrails

Systems & Data: Linux, Git, Python Multiprocessing, FastAPI, MLflow, Grafana, Pandas, NumPy, scikit-learn, XGBoost

Certifications: Microsoft AZ-900 Azure Fundamentals, Microsoft Azure Data Engineering, AWS Academy Cloud Foundations, DeepLearning.AI (Neural Networks & Deep Learning, AI Agents in LangGraph, RLHF)